

METARADAR SYSTEM ARCHITECTURE OVERVIEW

High-Level Data Flow & Technology Stack · Aura Pharmers (Novo Nordisk GBS Hackathon 2026)

EXTERNAL SIGNAL SOURCES

PubMed / Literature

ClinicalTrials.gov

Pharma News / Press

Congresses (ASH/ISTH)

Regulatory / Safety

Patient / Access

Synthetic Demo Data

LANGGRAPH INTELLIGENCE WORKFLOW — 9 STAGES

1-2. COLLECT & VALIDATE

Modular source connectors & PII/PHI filter

3. UNDERSTAND THE SIGNAL

Entity extraction & Haemophilia domain mapping

4. CONNECT RELATED EVIDENCE

Multi-source confluence & asset lifecycle tracking

5. CHALLENGE WEAK CONCLUSIONS

AI checks conflicting trial/cohort evidence

6-7. PRIORITIZE & ROUTE

Expected milestone watch & role relevance routing

8-9. EXPLAIN & LEARN

Fact/Interpretation tagging & calibration loop

AI REASONING LAYER: Local Gemma 3 4B GPU · Optional Grok API · BART / Extractive Summarization Fallback

Source-linked by design, with 100% source-linking as a prototype evaluation target.

DECISION-FIRST USER INTERFACES

Signal Radar

Priority signal feed

Four-Question Cards

Q1: What? Q2: Why? Q3: Route Q4: Act

Intelligence Digest

Role-filtered executive briefs

Ask Athena

Grounded conversational Q&A